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== Vitis HLS Report for 'matVecMul'

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* Date: Fri Jan 1 00:00:38 2021

* Version: 2021.1 (Build 3247384 on Thu Jun 10 19:36:07 MDT 2021)
* Project: matVecMul_stream_prj
* Solution: solution_fullunroll (Vivado IP Flow Target)
* Product family: zynq
* Target device: xc7z020-clg400-1

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== Performance Estimates

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+ Timing:

* Summary:

Clock	Target	Estimated	Uncertainty
ap_clk	10.00 ns	6.912 ns	2.70 ns

+ Latency:

* Summary:

Latency (cycles)		Latency (absolute)		Interval		Pipeline
min	max	min	max	min	max	Type
8903	179321	89.030 us	1.793 ms	8904	179322	no

+ Detail:

* Instance:

Instance	Module	Latency (cycles)		Latency (absolute)		Interval		Pipeline
		min	max	min	max	min	max	Type
grp_matVecMul_Pipeline_sample_loop_fu_1694	matVecMul_Pipeline_sample_loop	66	66	0.660 us	0.660 us	66	66	no
grp_matVecMul_Pipeline_matmul_inner_loop_fu_1841	matVecMul_Pipeline_matmul_inner_loop	130	131	1.300 us	1.310 us	130	131	no

* Loop:

Loop Name	Latency (cycles)		Iteration	Initiation	Interval	Trip	Pipelined
	min	max	Latency	achieved	target	Count	
- vector_loop	8902	179320	8902 ~ 8966	-	-	1 ~ 20	no
+ matmul_outer_loop	8832	8896	138 ~ 139	-	-	64	no

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== Utilization Estimates

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* Summary:

Name	BRAM_18K	DSP	FF	LUT	URAM
DSP	-	-	-	-	-
Expression	-	-	0	407	-
FIFO	-	-	-	-	-
Instance	128	192	31008	37050	-
Memory	-	-	-	-	-

Multiplexer	-	-	-	115	-
Register	-	-	9087	-	-
+-----+-----+-----+-----+-----+-----+					
Total	128	192	40095	37572	0
+-----+-----+-----+-----+-----+-----+					
Available	280	220	106400	53200	0
+-----+-----+-----+-----+-----+-----+					
Utilization (%)	45	87	37	70	0
+-----+-----+-----+-----+-----+-----+					

+ Detail:

* Instance:

Instance	Module	BRAM_18K	DSP	FF	LUT	URAM
control_s_axi_U	control_s_axi	128	0	5270	4712	0
grp_matVecMul_Pipeline_matmul_inner_loop_fu_1841	matVecMul_Pipeline_matmul_inner_loop	0	189	25539	31918	0
grp_matVecMul_Pipeline_sample_loop_fu_1694	matVecMul_Pipeline_sample_loop	0	0	34	97	0
mul_32s_32s_64_2_1_U331	mul_32s_32s_64_2_1	0	3	165	50	0
mux_646_70_1_1_U332	mux_646_70_1_1	0	0	0	273	0
Total		128	192	31008	37050	0

* DSP:

N/A

* Memory:

N/A

* FIFO:

N/A

* Expression:

Variable Name	Operation	DSP	FF	LUT	Bitwidth P0	Bitwidth P1
add_ln44_fu_2147_p2	+	0	0	39	32	1
i_2_fu_2131_p2	+	0	0	38	31	1
p_Val2_1_fu_2859_p2	+	0	0	39	32	32
sub22_fu_2108_p2	+	0	0	39	32	2
sub_fu_2102_p2	+	0	0	39	32	2
and_ln789_fu_2941_p2	and	0	0	2	1	1
and_ln790_fu_2952_p2	and	0	0	2	1	1
and_ln795_fu_2985_p2	and	0	0	2	1	1
carry_1_fu_2879_p2	and	0	0	2	1	1
out_stream_TLAST_int_regslice	and	0	0	2	1	1
overflow_fu_2979_p2	and	0	0	2	1	1
underflow_fu_2996_p2	and	0	0	2	1	1
Range1_all_ones_fu_2911_p2	icmp	0	0	14	22	2
Range1_all_zeros_fu_2917_p2	icmp	0	0	14	22	1
Range2_all_ones_fu_2895_p2	icmp	0	0	14	21	2
cmp21_fu_2137_p2	icmp	0	0	18	32	32
icmp_ln32_fu_2126_p2	icmp	0	0	18	32	32
icmp_ln44_fu_2142_p2	icmp	0	0	18	32	32
icmp_ln69_fu_3024_p2	icmp	0	0	18	32	32
or_ln384_fu_3010_p2	or	0	0	2	1	1
or_ln794_fu_2969_p2	or	0	0	2	1	1
deleted_ones_fu_2946_p3	select	0	0	2	1	1
deleted_zeros_fu_2923_p3	select	0	0	2	1	1
out_stream_TDATA_int_regslice	select	0	0	32	1	32
select_ln384_fu_3002_p3	select	0	0	33	1	31
xor_ln416_fu_2873_p2	xor	0	0	2	1	2
xor_ln789_fu_2935_p2	xor	0	0	2	1	2
xor_ln790_fu_2956_p2	xor	0	0	2	2	1
xor_ln794_1_fu_2974_p2	xor	0	0	2	1	2
xor_ln794_fu_2963_p2	xor	0	0	2	1	2

xor_ln795_fu_2990_p2	xor	0	0	2	2	1
Total		0	0	407	373	255

* Multiplexer:

Name	LUT	Input Size	Bits	Total Bits
ap_NS_fsm	65	13	1	13
ap_phi_mux_neg_src_1_phi_fu_1687_p4	14	3	1	3
i_fu_532	9	2	31	62
in_stream_TREADY_int_regslice	9	2	1	2
k_reg_1672	9	2	32	64
out_stream_TDATA_blk_n	9	2	1	2
Total	115	24	67	146

* Register:

Name	FF	LUT	Bits	Const Bits
A_0_load_reg_3876	32	0	32	0
A_10_load_reg_3926	32	0	32	0
A_11_load_reg_3931	32	0	32	0
A_12_load_reg_3936	32	0	32	0
A_13_load_reg_3941	32	0	32	0
A_14_load_reg_3946	32	0	32	0
A_15_load_reg_3951	32	0	32	0
A_16_load_reg_3956	32	0	32	0
A_17_load_reg_3961	32	0	32	0
A_18_load_reg_3966	32	0	32	0
A_19_load_reg_3971	32	0	32	0
A_1_load_reg_3881	32	0	32	0
A_20_load_reg_3976	32	0	32	0
A_21_load_reg_3981	32	0	32	0
A_22_load_reg_3986	32	0	32	0
A_23_load_reg_3991	32	0	32	0
A_24_load_reg_3996	32	0	32	0
A_25_load_reg_4001	32	0	32	0
A_26_load_reg_4006	32	0	32	0
A_27_load_reg_4011	32	0	32	0
A_28_load_reg_4016	32	0	32	0
A_29_load_reg_4021	32	0	32	0
A_2_load_reg_3886	32	0	32	0
A_30_load_reg_4026	32	0	32	0
A_31_load_reg_4031	32	0	32	0
A_32_load_reg_4036	32	0	32	0
A_33_load_reg_4041	32	0	32	0
A_34_load_reg_4046	32	0	32	0
A_35_load_reg_4051	32	0	32	0
A_36_load_reg_4056	32	0	32	0
A_37_load_reg_4061	32	0	32	0
A_38_load_reg_4066	32	0	32	0
A_39_load_reg_4071	32	0	32	0
A_3_load_reg_3891	32	0	32	0
A_40_load_reg_4076	32	0	32	0
A_41_load_reg_4081	32	0	32	0
A_42_load_reg_4086	32	0	32	0
A_43_load_reg_4091	32	0	32	0
A_44_load_reg_4096	32	0	32	0
A_45_load_reg_4101	32	0	32	0
A_46_load_reg_4106	32	0	32	0
A_47_load_reg_4111	32	0	32	0
A_48_load_reg_4116	32	0	32	0
A_49_load_reg_4121	32	0	32	0

A_4_load_reg_3896	32	0	32	0
A_50_load_reg_4126	32	0	32	0
A_51_load_reg_4131	32	0	32	0
A_52_load_reg_4136	32	0	32	0
A_53_load_reg_4141	32	0	32	0
A_54_load_reg_4146	32	0	32	0
A_55_load_reg_4151	32	0	32	0
A_56_load_reg_4156	32	0	32	0
A_57_load_reg_4161	32	0	32	0
A_58_load_reg_4166	32	0	32	0
A_59_load_reg_4171	32	0	32	0
A_6_load_reg_3901	32	0	32	0
A_60_load_reg_4176	32	0	32	0
A_61_load_reg_4181	32	0	32	0
A_62_load_reg_4186	32	0	32	0
A_63_load_reg_3545	32	0	32	0
A_6_load_reg_3906	32	0	32	0
A_7_load_reg_3911	32	0	32	0
A_8_load_reg_3916	32	0	32	0
A_9_load_reg_3921	32	0	32	0
Range1_all_ones_reg_4424	1	0	1	0
Range1_all_zeros_reg_4431	1	0	1	0
Range2_all_ones_reg_4419	1	0	1	0
acc_V_0_loc_fu_788	70	0	70	0
acc_V_lcssa_reg_4391	70	0	70	0
add_ln44_reg_3468	32	0	32	0
ap_CS_fsm	12	0	12	0
carry_1_reg_4412	1	0	1	0
cmp21_reg_3460	1	0	1	0
col_size_read_reg_3042	32	0	32	0
grp_matVecMul_Pipeline_matmul_inner_loop_fu_1841_ap_start_reg	1	0	1	0
grp_matVecMul_Pipeline_sample_loop_fu_1694_ap_start_reg	1	0	1	0
i_2_reg_3455	31	0	31	0
i_fu_532	31	0	31	0
k_reg_1672	32	0	32	0
mul_ln1171_reg_4380	64	0	64	0
no_of_vectors_read_reg_3053	32	0	32	0
p_Result_2_reg_4406	1	0	1	0
p_Result_s_reg_4396	1	0	1	0
p_Val2_1_reg_4401	32	0	32	0
row_size_read_reg_3048	32	0	32	0
select_ln340_10_loc_fu_576	70	0	70	0
select_ln340_11_loc_fu_580	70	0	70	0
select_ln340_12_loc_fu_584	70	0	70	0
select_ln340_13_loc_fu_588	70	0	70	0
select_ln340_14_loc_fu_592	70	0	70	0
select_ln340_15_loc_fu_596	70	0	70	0
select_ln340_16_loc_fu_600	70	0	70	0
select_ln340_17_loc_fu_604	70	0	70	0
select_ln340_18_loc_fu_608	70	0	70	0
select_ln340_19_loc_fu_612	70	0	70	0
select_ln340_1_loc_fu_540	70	0	70	0
select_ln340_20_loc_fu_616	70	0	70	0
select_ln340_21_loc_fu_620	70	0	70	0
select_ln340_22_loc_fu_624	70	0	70	0
select_ln340_23_loc_fu_628	70	0	70	0
select_ln340_24_loc_fu_632	70	0	70	0
select_ln340_25_loc_fu_636	70	0	70	0
select_ln340_26_loc_fu_640	70	0	70	0
select_ln340_27_loc_fu_644	70	0	70	0
select_ln340_28_loc_fu_648	70	0	70	0
select_ln340_29_loc_fu_652	70	0	70	0
select_ln340_2_loc_fu_544	70	0	70	0
select_ln340_30_loc_fu_656	70	0	70	0
select_ln340_31_loc_fu_660	70	0	70	0
select_ln340_32_loc_fu_664	70	0	70	0
select_ln340_33_loc_fu_668	70	0	70	0

select_ln340_34_loc_fu_672	70	0	70	0
select_ln340_35_loc_fu_676	70	0	70	0
select_ln340_36_loc_fu_680	70	0	70	0
select_ln340_37_loc_fu_684	70	0	70	0
select_ln340_38_loc_fu_688	70	0	70	0
select_ln340_39_loc_fu_692	70	0	70	0
select_ln340_3_loc_fu_548	70	0	70	0
select_ln340_40_loc_fu_696	70	0	70	0
select_ln340_41_loc_fu_700	70	0	70	0
select_ln340_42_loc_fu_704	70	0	70	0
select_ln340_43_loc_fu_708	70	0	70	0
select_ln340_44_loc_fu_712	70	0	70	0
select_ln340_45_loc_fu_716	70	0	70	0
select_ln340_46_loc_fu_720	70	0	70	0
select_ln340_47_loc_fu_724	70	0	70	0
select_ln340_48_loc_fu_728	70	0	70	0
select_ln340_49_loc_fu_732	70	0	70	0
select_ln340_4_loc_fu_552	70	0	70	0
select_ln340_50_loc_fu_736	70	0	70	0
select_ln340_51_loc_fu_740	70	0	70	0
select_ln340_52_loc_fu_744	70	0	70	0
select_ln340_53_loc_fu_748	70	0	70	0
select_ln340_54_loc_fu_752	70	0	70	0
select_ln340_55_loc_fu_756	70	0	70	0
select_ln340_56_loc_fu_760	70	0	70	0
select_ln340_57_loc_fu_764	70	0	70	0
select_ln340_58_loc_fu_768	70	0	70	0
select_ln340_59_loc_fu_772	70	0	70	0
select_ln340_5_loc_fu_556	70	0	70	0
select_ln340_60_loc_fu_776	70	0	70	0
select_ln340_61_loc_fu_780	70	0	70	0
select_ln340_62_loc_fu_784	70	0	70	0
select_ln340_6_loc_fu_560	70	0	70	0
select_ln340_7_loc_fu_564	70	0	70	0
select_ln340_8_loc_fu_568	70	0	70	0
select_ln340_9_loc_fu_572	70	0	70	0
select_ln340_loc_fu_536	70	0	70	0
sub22_reg_3447	32	0	32	0
sub_reg_3442	32	0	32	0
targetBlock_reg_4386	6	0	6	0
x_localbuff_V_0	32	0	32	0
x_localbuff_V_1	32	0	32	0
x_localbuff_V_10	32	0	32	0
x_localbuff_V_11	32	0	32	0
x_localbuff_V_12	32	0	32	0
x_localbuff_V_13	32	0	32	0
x_localbuff_V_14	32	0	32	0
x_localbuff_V_15	32	0	32	0
x_localbuff_V_16	32	0	32	0
x_localbuff_V_17	32	0	32	0
x_localbuff_V_18	32	0	32	0
x_localbuff_V_19	32	0	32	0
x_localbuff_V_2	32	0	32	0
x_localbuff_V_20	32	0	32	0
x_localbuff_V_21	32	0	32	0
x_localbuff_V_22	32	0	32	0
x_localbuff_V_23	32	0	32	0
x_localbuff_V_24	32	0	32	0
x_localbuff_V_25	32	0	32	0
x_localbuff_V_26	32	0	32	0
x_localbuff_V_27	32	0	32	0
x_localbuff_V_28	32	0	32	0
x_localbuff_V_29	32	0	32	0
x_localbuff_V_3	32	0	32	0
x_localbuff_V_30	32	0	32	0
x_localbuff_V_31	32	0	32	0
x_localbuff_V_32	32	0	32	0

x_localbuff_V_33	32	0	32	0
x_localbuff_V_34	32	0	32	0
x_localbuff_V_35	32	0	32	0
x_localbuff_V_36	32	0	32	0
x_localbuff_V_37	32	0	32	0
x_localbuff_V_38	32	0	32	0
x_localbuff_V_39	32	0	32	0
x_localbuff_V_4	32	0	32	0
x_localbuff_V_40	32	0	32	0
x_localbuff_V_41	32	0	32	0
x_localbuff_V_42	32	0	32	0
x_localbuff_V_43	32	0	32	0
x_localbuff_V_44	32	0	32	0
x_localbuff_V_45	32	0	32	0
x_localbuff_V_46	32	0	32	0
x_localbuff_V_47	32	0	32	0
x_localbuff_V_48	32	0	32	0
x_localbuff_V_49	32	0	32	0
x_localbuff_V_5	32	0	32	0
x_localbuff_V_50	32	0	32	0
x_localbuff_V_51	32	0	32	0
x_localbuff_V_52	32	0	32	0
x_localbuff_V_53	32	0	32	0
x_localbuff_V_54	32	0	32	0
x_localbuff_V_55	32	0	32	0
x_localbuff_V_56	32	0	32	0
x_localbuff_V_57	32	0	32	0
x_localbuff_V_58	32	0	32	0
x_localbuff_V_59	32	0	32	0
x_localbuff_V_6	32	0	32	0
x_localbuff_V_60	32	0	32	0
x_localbuff_V_61	32	0	32	0
x_localbuff_V_62	32	0	32	0
x_localbuff_V_63	32	0	32	0
x_localbuff_V_7	32	0	32	0
x_localbuff_V_8	32	0	32	0
x_localbuff_V_9	32	0	32	0
zext_ln44_reg_3473	32	0	64	32
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Total	9087	0	9119	32
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== Interface
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* Summary:
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RTL Ports	Dir	Bits	Protocol	Source Object	C Type
s_axi_control_AWVALID	in	1	s_axi	control	array
s_axi_control_AWREADY	out	1	s_axi	control	array
s_axi_control_AWADDR	in	15	s_axi	control	array
s_axi_control_WVALID	in	1	s_axi	control	array
s_axi_control_WREADY	out	1	s_axi	control	array
s_axi_control_WDATA	in	32	s_axi	control	array
s_axi_control_WSTRB	in	4	s_axi	control	array
s_axi_control_ARVALID	in	1	s_axi	control	array
s_axi_control_ARREADY	out	1	s_axi	control	array
s_axi_control_ARADDR	in	15	s_axi	control	array
s_axi_control_RVALID	out	1	s_axi	control	array
s_axi_control_RREADY	in	1	s_axi	control	array
s_axi_control_RDATA	out	32	s_axi	control	array
s_axi_control_RRESP	out	2	s_axi	control	array
s_axi_control_BVALID	out	1	s_axi	control	array
s_axi_control_BREADY	in	1	s_axi	control	array

s_axi_control_BRESP	out	2	s_axi	control	array
ap_clk	in	1	ap_ctrl_hs	matVecMul	return value
ap_rst_n	in	1	ap_ctrl_hs	matVecMul	return value
interrupt	out	1	ap_ctrl_hs	matVecMul	return value
in_stream_TDATA	in	32	axis	in_stream_V_data_V	pointer
in_stream_TVALID	in	1	axis	in_stream_V_dest_V	pointer
in_stream_TREADY	out	1	axis	in_stream_V_dest_V	pointer
in_stream_TDEST	in	1	axis	in_stream_V_dest_V	pointer
in_stream_TKEEP	in	4	axis	in_stream_V_keep_V	pointer
in_stream_TSTRB	in	4	axis	in_stream_V_strb_V	pointer
in_stream_TUSER	in	1	axis	in_stream_V_user_V	pointer
in_stream_TLAST	in	1	axis	in_stream_V_last_V	pointer
in_stream_TID	in	1	axis	in_stream_V_id_V	pointer
out_stream_TDATA	out	32	axis	out_stream_V_data_V	pointer
out_stream_TVALID	out	1	axis	out_stream_V_dest_V	pointer
out_stream_TREADY	in	1	axis	out_stream_V_dest_V	pointer
out_stream_TDEST	out	1	axis	out_stream_V_dest_V	pointer
out_stream_TKEEP	out	4	axis	out_stream_V_keep_V	pointer
out_stream_TSTRB	out	4	axis	out_stream_V_strb_V	pointer
out_stream_TUSER	out	1	axis	out_stream_V_user_V	pointer
out_stream_TLAST	out	1	axis	out_stream_V_last_V	pointer
out_stream_TID	out	1	axis	out_stream_V_id_V	pointer